

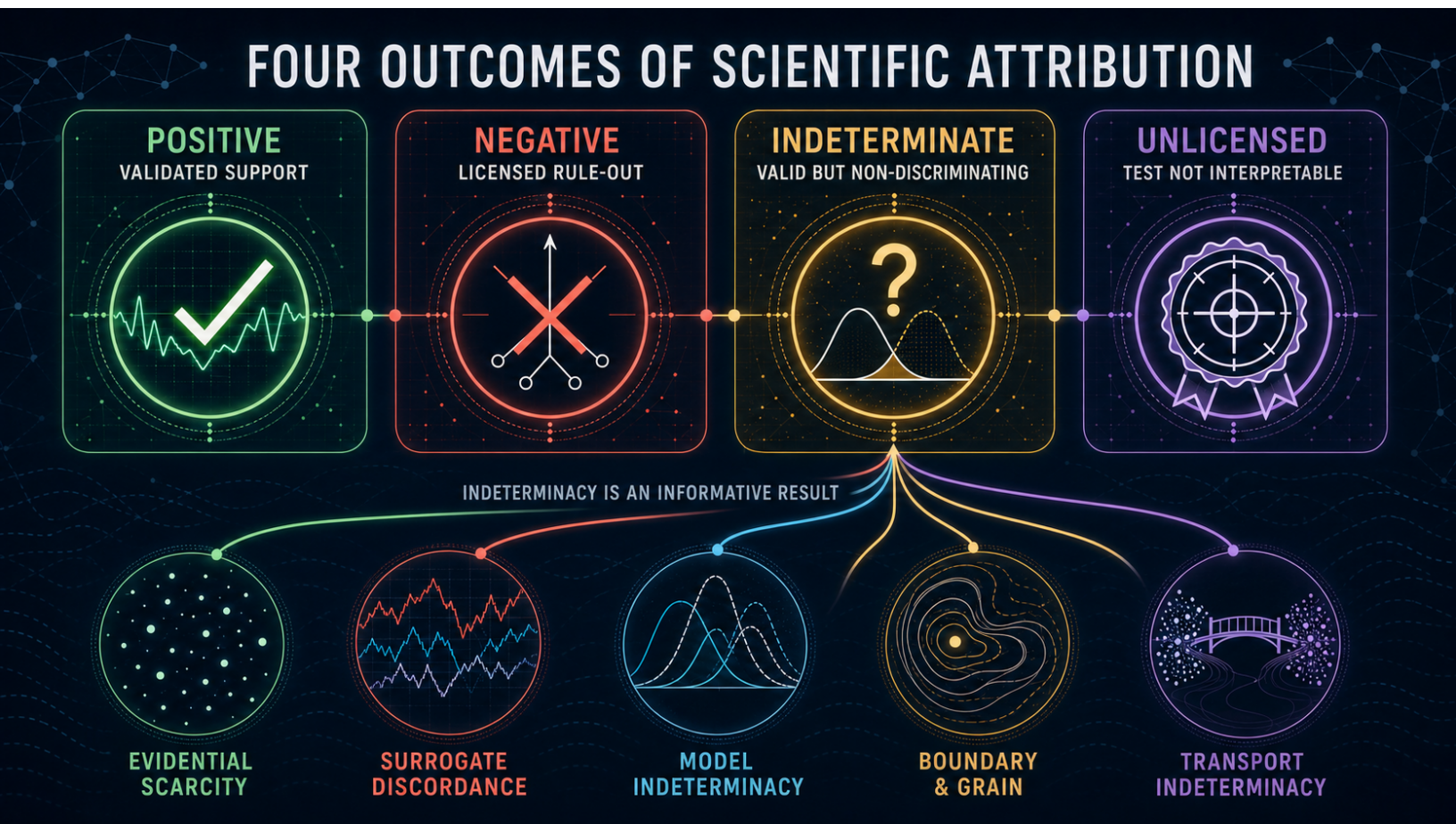
Indeterminacy as a Scientific Result

A Four-Outcome Framework for Consciousness Attribution

Phil Stilwell

Independent Scholar
ORCID: 0009-0009-3725-8682

A visual preview:



Abstract

Tests for consciousness are increasingly applied where ordinary verbal report is absent, unreliable, or inapplicable: severe brain injury, infancy, nonhuman animals, neural organoids, and artificial systems. In these cases a negative test is often treated as evidence that consciousness is absent, while conflicting or non-transferable evidence is compressed into a vague category of uncertainty. Diagnostic research, three-way decision theory, and classifiers with a reject option already show why forced binary output can be misleading. Consciousness attribution adds a further problem: an assessment may be insufficiently discriminating even though valid, or it may lack a licensed interpretation altogether. This paper therefore proposes four outputs: positive, negative, indeterminate, and unlicensed. A result is indeterminate when an applicable and adequately executed assessment yields evidence that does not discriminate between presence and absence. An assessment is unlicensed when its validity conditions are not met, its target or system is inadequately specified, or its evidential interpretation has not been transported to the population or substrate. The formal rule gives validity failure precedence over evidential thresholds and gives unresolved surrogate conflict precedence over scalar aggregation. Indeterminate and unlicensed outputs additionally receive a cause profile identifying evidential scarcity, surrogate discordance, model disagreement, boundary instability, or transport failure. The framework links each unresolved output to a specific research response and separates scientific classification from precautionary action. It applies across competing theories and can be incorporated into theory-driven and theory-light consciousness tests.

Keywords: consciousness attribution; indeterminacy; C-tests; diagnostic uncertainty; disorders of consciousness; animal consciousness; artificial consciousness; precaution

1. Introduction

The easiest consciousness experiments begin with subjects who can understand instructions, report their experiences, and repeat the task. The hardest cases are precisely those in which these conveniences disappear. A patient may retain awareness while lacking reliable motor output. An infant or nonhuman animal may be unable to understand the reporting protocol. An organoid may have no behavior rich enough to interpret. An artificial system may reproduce human-like reports while differing radically from the systems in which those reports were calibrated. These cases require tests for consciousness, or C-tests, that do not simply presuppose ordinary adult human report (Bayne et al., 2024).

The scientific difficulty is not exhausted by imperfect instruments. Consciousness science lacks a universally accepted theory, a generally applicable gold standard, and a single measurement vocabulary that transfers cleanly across populations and substrates (Bayne et al., 2024; Michel et al., 2019; Seth & Bayne, 2022). Behavioral, computational, anatomical, electrophysiological, perturbational, and theory-derived indicators can converge, diverge, or fail to apply. Yet the public and scientific language surrounding these assessments remains strongly binary. A system is described as conscious or nonconscious; a test is said to detect or fail to detect consciousness; an indicator is present or absent. Results that do not fit these categories are often described merely as uncertain, excluded from analysis, or absorbed into the negative class.

This paper argues that indeterminacy should be treated as a scientific result rather than as an embarrassment awaiting deletion. The claim is not that consciousness itself must be metaphysically indeterminate. The claim concerns warranted attribution: given a specified target, system, test, evidence base, and validity envelope, the evidence may fail to license either a positive or a negative classification. That failure can be informative. It

may identify inadequate sensitivity, disagreement among credible surrogates, dependence on a contested theory, instability in the boundary of the candidate system, or failed transfer from the population in which the test was validated.

The proposal has two parts. First, consciousness assessments should return four possible outcomes:

1. **Positive:** the evidence licenses attribution of the target form of consciousness.
2. **Negative:** the evidence licenses denial of the target form of consciousness.
3. **Indeterminate:** the assessment is applicable and adequately executed, but the evidence does not discriminate sufficiently between presence and absence.
4. **Unlicensed:** one or more conditions required to interpret the assessment are not satisfied.

Second, indeterminate and unlicensed outputs should be accompanied by a cause profile. A bare label such as “uncertain” does little scientific work. By contrast, identifying a result as indeterminate because validated measures conflict suggests replication or adjudication; identifying it as unlicensed because the test has not been transported across substrates suggests a calibration program; identifying it as boundary-sensitive suggests interventions on candidate system boundaries.

This structure is intended to be theory-plural rather than theory-neutral in the strong sense. No consciousness test is wholly independent of assumptions about its target and evidential relevance. The framework instead requires those assumptions to be stated and prevents the output of one test from silently acquiring a wider interpretation than its validation permits.

1.1 Method and contribution

This is a methodological and philosophy-of-science paper. It does not report a new empirical dataset or claim that the proposed thresholds have already been calibrated. The argument proceeds by reconstructing the inferential steps that any consciousness assessment must take, comparing those steps with practices in diagnostic testing and abstaining classification, and testing the resulting framework against difficult applications.

The proposal is evaluated by four design requirements:

1. **Non-collapse:** absence of a positive result must not automatically become a negative result, and failed applicability must not be pooled with valid nondiscrimination.
2. **Auditability:** targets, system boundaries, auxiliary assumptions, validity conditions, thresholds, and conflict rules must be open to inspection.
3. **Resolution specificity:** every unresolved output must identify what kind of evidence or validation work could change it.
4. **Domain discipline:** results may not travel across populations, species, tasks, or substrates without an explicit transport argument.

The paper does not claim priority for abstention, gray-zone diagnosis, or three-way decisions. Its contribution is the consciousness-specific synthesis: a four-output scheme in which indeterminate is an evidential result, unlicensed is a meta-assessment of inferential failure, and both receive cause profiles that direct different resolution procedures. The appendices turn this synthesis into a protocol, reporting checklist, and misuse audit suitable for preregistration or test validation.

2. Why Binary Classification Fails

2.1 Consciousness tests lack a universal gold standard

Diagnostic science ordinarily evaluates a test against an accepted reference standard. For consciousness, the most secure reference cases are awake, communicative adult humans, but the very need for C-tests arises as systems become less similar to those cases. Even in disorders of consciousness, behavioral examination, task-based neuroimaging, resting-state measures, and perturbational indices have different cognitive demands and different routes to false negatives. Peterson et al. (2016) argue that standard validation is especially difficult where no single measure functions as an adequate gold standard and recommend consilience across partially independent evidence.

The problem is not solved by calling report the gold standard. Report is powerful evidence when it is credible, but failure to report can result from impaired language comprehension, memory, attention, motivation, arousal, or motor output. No-report paradigms were developed partly because neural activity associated with report preparation can be mistaken for activity constitutive of experience (Tsuchiya et al., 2015). In patients with severe brain injury, repeated behavioral assessments improve diagnostic reliability (Wannez et al., 2017), and task-based fMRI or EEG can reveal cognitive motor dissociation in a substantial subset of behaviorally unresponsive patients (Bodien et al., 2024). These findings do not establish that every unresponsive patient is conscious. They establish that behavioral non-response is not, by itself, strong evidence of absence.

2.2 The cost of a false negative is easily hidden

A positive consciousness result usually demands explanation. Investigators ask whether the signal reflects artifact, unconscious processing, task compliance, or overfitting. Negative results often receive less scrutiny. But a negative output is warranted only if the test had a reasonable opportunity to detect the target. A patient who fails a demanding imagery task may lack the attention, comprehension, memory, or sustained control required by the task while retaining experience. An animal may fail a human-designed paradigm for ecological rather than phenomenological reasons. An AI system may fail a biological marker because the marker is not substrate-general. An organoid may produce signals that resemble a human neural correlate without satisfying the conditions under which that correlate was validated.

The general diagnostic literature distinguishes valid intermediate results from uninterpretable or invalid ones and warns that excluding inconclusive outputs can inflate estimates of accuracy (Shinkins et al., 2013; Simel et al., 1987). Consciousness science needs the same discipline, with an additional complication: errors can arise not only from the test but also from disagreement about the target and the theory connecting evidence to that target.

2.3 Heterogeneous uncertainty is flattened

Consider three cases:

- A well-calibrated perturbational measure returns a value between preregistered rule-in and rule-out thresholds.
- Two independently validated indicators return opposed results.
- A measure validated in adult mammalian brains is applied to a transformer-based artificial system.

All three may be called uncertain, but they are not scientifically equivalent. The first is a valid gray-zone result. The second is unresolved evidential discordance. The third may not be an interpretable consciousness result at all. It is an extrapolation whose validity remains to be established. Pooling these cases conceals what is known, what failed, and what investigation should occur next.

2.4 Disagreement among theories changes the evidential map

Leading theories of consciousness emphasize different properties: integration, global availability, recurrent processing, higher-order representation, predictive organization, or other dynamical and representational features (Seth & Bayne, 2022). Theory-derived indicator approaches can make this disagreement productive by asking whether a system possesses properties associated with several theories (Butlin et al., 2023). But indicator accumulation does not remove model uncertainty. If one theory treats a feature as necessary while another treats it as incidental, the same evidence can support incompatible classifications.

This matters especially beyond familiar biological cases. Similarity to a known conscious system weakens as anatomy, development, embodiment, behavior, and causal organization diverge. A result may therefore be empirically precise while remaining interpretively unstable. Indeterminacy is sometimes located not in the data but in the bridge from data to consciousness.

2.5 Four distinctions that the framework preserves

The argument depends on keeping four questions apart. Their answers may interact, but no answer should silently substitute for another.

First, **the state of the world is not the state of the evidence**. A system may in fact be conscious or nonconscious even when available evidence does not warrant either classification. Epistemic indeterminacy therefore does not entail that consciousness itself has an intermediate ontological status. It marks a limitation in what a specified assessment licenses.

Second, **the target is not the indicator**. Behavioral command following, perturbational complexity, recurrent activity, global availability, and self-report can provide evidence for consciousness under appropriate assumptions. None is identical to phenomenal consciousness merely because it functions as a test. This target-indicator distinction is what makes false positives, false negatives, and construct-validity questions intelligible.

Third, **non-detection is not negative evidence by default**. A result can fail to cross a positive threshold because the target is absent, because the measure is insensitive, because an auxiliary capacity is impaired, because the system was measured at the wrong grain, or because the test does not transfer to the case. Only some of these possibilities support a negative attribution.

Fourth, **scientific classification is not practical action**. A scientific result describes what the evidence warrants. A clinical, ethical, or governance decision additionally depends on stakes, reversibility, available alternatives, and the costs of error. An indeterminate result can therefore support precaution without being relabeled positive, and it can support continued investigation without being relabeled negative.

These distinctions constrain the formalism that follows. The assessment tuple keeps target, system, evidence, and validity separate. The four-outcome rule distinguishes nondiscrimination from non-applicability. The cause

profile explains why a result remains unresolved. The action rule then takes scientific status as one input among others rather than disguising a decision as a discovery.

2.6 Relation to diagnostic gray zones and abstaining classifiers

The rejection of forced binary output has substantial precedent. Classifiers with a reject option can abstain when classification is insufficiently reliable, trading coverage for a lower conditional error rate (Chow, 1970). Three-way decision theory similarly distinguishes acceptance, rejection, and a boundary region of noncommitment (Yao, 2010). Diagnostic methodology distinguishes valid inconclusive results from invalid or uninterpretable assessments and warns that excluding them can distort apparent accuracy (Shinkins et al., 2013). STARD reporting guidance requires transparent reporting of participant flow, test execution, indeterminate results, and exclusions so that bias and applicability can be assessed (Bossuyt et al., 2015).

The present proposal inherits the lesson that abstention can be epistemically superior to forced classification, but it does not simply rename a reject option. A reject option usually presupposes that the classifier is applicable to the case and then declines to classify because confidence is low. That corresponds most closely to **indeterminate**. Consciousness assessment also needs a distinct status for cases in which the score lacks the intended interpretation - for example, because the target was unspecified, data quality failed, the candidate system was not bounded, or a human-calibrated measure was applied to an artificial substrate without bridge validation. That is **unlicensed**.

The distinction can be summarized as follows:

Question	If no	Output
Does this assessment have an established interpretation for this target and system?	The inference cannot yet be made.	Unlicensed
Given a licensed assessment, does the evidence discriminate beyond preregistered conflict and threshold rules?	The evidence bears on the question but does not settle it.	Indeterminate

This additional layer is important in consciousness science because the difficult cases are often not merely near a familiar decision boundary. They are outside the domains in which the evidence-target relation was established. The framework therefore combines abstention with an explicit applicability audit.

3. Four Outcomes, Not Two

3.1 The assessment tuple

A consciousness attribution should be indexed to at least five elements:

$$\mathcal{A} = \langle S, C, T, E, M, \mathcal{V} \rangle,$$

where:

- S is the candidate system at a specified spatial and temporal grain;

- C is the target, such as phenomenal presence, a particular conscious content, wakeful awareness, sentience, or self-consciousness;
- T is the test or assessment battery;
- E is the evidence produced;
- M is the inferential model connecting the evidence to the target;
- $\mathcal{V}$ is the validity envelope within which the test has an established interpretation.

The validity envelope includes the populations, substrates, task conditions, recording quality, system boundaries, and auxiliary capacities for which the inferential relation between E and C has been justified. This follows a general measurement lesson: validity concerns the interpretation and target relation of a measure, not merely its reliability or numerical reproducibility (Borsboom et al., 2004). A result is not simply “positive for consciousness.” It is positive for a specified target, in a specified system, under an assessment whose scope is explicit.

Plain-language reading. A consciousness result is not a property of a score in isolation. It is a claim that this test, applied to this bounded system, produced this evidence about this form of consciousness under an explicit inferential model and conditions in which the test has an established meaning.

Why this matters. The tuple blocks inferential drift. Evidence for command comprehension cannot automatically become evidence for continuous phenomenal richness; a cortical measure cannot automatically classify a whole brain-body system; and a test calibrated in responsive adults cannot automatically retain its interpretation in an organoid or artificial architecture.

Hidden assumptions being exposed. Every application presupposes a candidate system boundary, a target construct, a model of evidential relevance, auxiliary capacities, and a transport claim from validation cases to the present case. Writing these elements into $\mathcal{A}$ makes them available for criticism rather than leaving them in the background.

What would show that the tuple is inadequate. The tuple should be expanded if two assessments with identical values for S , C , T , E , M , and $\mathcal{V}$ can nevertheless warrant different classifications because of a further scientifically relevant factor. Candidate additions might include an explicit decision context or time-indexed history. The present six-place form is proposed as a minimum reporting structure, not as a metaphysically complete analysis of attribution.

3.2 A four-outcome decision rule

Let $a(\mathcal{A})$ indicate whether the test has a licensed interpretation for the specified system, target, model, and validity envelope. Let $q(\mathcal{A})$ indicate whether execution and data quality meet preregistered requirements. Let $c(E, M)$ flag unresolved conflict among credible surrogates under the inferential model. Finally, let $L(E; T, C, M)$ represent the degree of evidential support supplied by a validated aggregation rule, with a rule-out threshold τ_- and rule-in threshold τ_+ , where $\tau_- < \tau_+$. Then:

$$D(\mathcal{A}) = \begin{cases} \text{Unlicensed,} & a = 0 \text{ or } q = 0 \\ \text{Indeterminate,} & a = 1, q = 1, c = 1 \\ \text{Positive,} & a = 1, q = 1, c = 0, L \geq \tau_+ \\ \text{Negative,} & a = 1, q = 1, c = 0, L \leq \tau_- \\ \text{Indeterminate,} & a = 1, q = 1, c = 0, \tau_- < L < \tau_+. \end{cases}$$

This representation is schematic. L could be a likelihood ratio, posterior probability, validated score, structured evidence grade, or result of a multi-indicator decision procedure. The framework does not require consensus on Bayesian or frequentist interpretation. It requires explicit thresholds, a preregistered conflict rule, and a distinction between valid nondiscrimination and a result that cannot bear the intended interpretation.

The order of the cases expresses a substantive precedence policy. Validity and quality are evaluated before evidential strength. Unresolved conflict is evaluated before scalar aggregation. Only an applicable, adequately executed, nonconflicted assessment reaches the threshold comparison. A battery therefore cannot select whichever surrogate yields the preferred answer after the result is known.

Plain-language reading. The test first has to qualify for use and produce adequate data. Only then should its evidential score be compared with a rule-in region, a rule-out region, and the interval between them. A qualified test can return a genuinely indeterminate result. A disqualified or failed test returns no licensed consciousness classification.

Role of the symbols. The applicability gate a concerns whether T , interpreted through M , bears a validated relation to C in system S . The quality gate q concerns whether the particular administration generated interpretable evidence. The conflict flag c prevents incompatible credible surrogates from being silently averaged. The score L represents evidential support, not an amount of consciousness. The thresholds τ_- and τ_+ are decision boundaries whose justification depends on validation and acceptable error rates; they are not assumed to mark natural joints in phenomenology.

Why this matters. A conventional binary classifier often uses one cutoff, so every observation below it is treated as negative. The two-threshold structure creates a protected region in which the evidence is valid but insufficiently discriminating. The applicability and quality gates prevent invalid assessments from entering even that gray zone, while the conflict gate preserves informative disagreement.

Assumptions and scope. The compact rule displays scalar L , but many consciousness batteries will be multivariate. Reduction to one score is legitimate only when the combination rule is validated and preregistered. Otherwise the component results and their conflicts must remain visible. Likewise, a and q may be graded in practice; their binary representation here identifies logical gates rather than requiring crude measurement.

What would count against the rule. The framework would require revision if validated C-tests cannot sustain separable rule-in and rule-out regions, if applicability cannot be assessed independently of the desired outcome, or if the indeterminate category systematically conceals results that reliably predict the target. Evidence of that kind might favor a fully probabilistic or multidimensional output. Such a revision would change the decision implementation while preserving the central distinction between nondiscrimination and unlicensed inference.

3.3 Positive

A positive result means that the evidence licenses attribution of the specified target under the stated model and validity envelope. It need not imply certainty, theoretical finality, or a complete explanation of experience. A positive command-following result in a behaviorally unresponsive patient can provide strong evidence of preserved awareness without revealing the character, continuity, or welfare significance of the patient’s experiences. A positive result is local to the target it tests.

Positive License Principle: Presence of a consciousness indicator licenses a positive attribution only when the indicator has validated relevance and adequate specificity for the target, the assessment is applicable and properly executed, and credible nonconscious or artifactual explanations have been controlled to the degree required by the rule-in standard.

Let I be a detected indicator and h_+ state that the positive-license conditions hold:

$$I \not\Rightarrow C, \quad h_+ \wedge I \Rightarrow_T \text{evidence for } C.$$

Plain-language reading. Finding an impressive marker is not enough. The marker supports consciousness only when investigators have justified what it measures, shown that it discriminates the target from relevant alternatives, and applied it within its validity envelope.

Why symmetry matters. The paper emphasizes false negatives because non-response is easily overread as absence, but false positives are also possible. Complex activity, flexible behavior, verbal self-ascription, or architectural resemblance may arise for reasons that do not support the specified consciousness target. Positive and negative classifications therefore both require licenses; neither is a default.

What would defeat a positive classification. Artifact, target ambiguity, low specificity, task confounding, circular validation, model failure, or unlicensed transport removes the positive license. As with a defeated negative classification, the result becomes indeterminate or unlicensed rather than automatically reversing polarity.

3.4 Negative

A negative result is stronger than failure to obtain a positive result. It asserts that the assessment was applicable, adequately sensitive, properly executed, and yielded evidence within a validated rule-out region. This paper proposes the following negative license principle:

Negative License Principle: Absence of a consciousness indicator licenses a negative consciousness attribution only when the indicator was expected to be detectable if the target were present, the test was applicable to the system, auxiliary task demands were satisfied or controlled, data quality was adequate, and plausible alternative causes of non-detection were excluded.

When these conditions are not satisfied, “not detected” should not be translated into “absent.” This principle does not make negative attribution impossible. It makes its evidential burden visible.

The prohibited inference and its licensed replacement can be displayed compactly. Let I be the indicator the test is designed to detect and let h state that detection conditions and auxiliary requirements are satisfied:

$$\neg I \not\Rightarrow \neg C,$$

but, for a test with validated rule-out performance,

$$h \wedge \neg I \Rightarrow_T \text{evidence for } \neg C.$$

Plain-language reading. Failure to observe an indicator does not by itself imply absence of consciousness. It becomes negative evidence only when the test should have displayed that indicator if the target were present and when the conditions needed for detection actually held.

Why the implication is indexed to T . The symbol $\Rightarrow_T$ denotes an inference licensed by a particular validated test, not a logical entailment. A negative EEG result, behavioral result, or architectural result can be strong evidence without making consciousness logically impossible. The subscript prevents empirical support from being mistaken for definition.

What would defeat a negative classification. Demonstrating inadequate sensitivity, unmet task demands, poor data quality, target mismatch, or a credible alternative mechanism for non-detection removes the negative license. It does not thereby establish consciousness; it changes the output to indeterminate or unlicensed, depending on whether the remaining assessment is interpretable.

3.5 Indeterminate

An indeterminate result occurs when the assessment genuinely bears on the question but does not discriminate enough to license either attribution or denial. Examples include scores in a validated gray zone, credible but conflicting indicators, and evidence that leaves multiple live explanatory models comparably supported.

Indeterminacy is therefore a result about the relation among target, evidence, and inferential standard. It is not a positive claim that the system occupies an ontologically intermediate state between consciousness and nonconsciousness. Consciousness may be binary, graded, multidimensional, or vague; the present framework is compatible with each possibility. It concerns what the assessment licenses.

3.6 Unlicensed

An unlicensed result occurs when the assessment cannot support the intended conclusion. Causes include poor signal quality, protocol failure, an unspecified target, an unstable system boundary, absent validation in the target population, or application outside the test's substrate domain. "Unlicensed" is preferable to "indeterminate" in these cases because the test has not produced a valid intermediate consciousness result. It has failed to establish an interpretable relation to the target.

The distinction matters statistically and rhetorically. If unlicensed cases are placed in the negative class, specificity may appear high while false negatives are hidden. If they are pooled with valid indeterminate cases, the test's limitations become harder to diagnose. Reporting them separately preserves the difference between a difficult case and an invalid inference.

Strictly speaking, unlicensed is a meta-output about the assessment rather than an evidential verdict about the system. Positive, negative, and indeterminate answer the target question at different levels of decisiveness. Unlicensed states that the present procedure cannot yet answer that question. The phrase “four-outcome framework” refers to the four outputs that a reporting system must return, not to four possible ontological states of consciousness.

This status should be local rather than contagious. One unlicensed measure does not invalidate every other measure in a battery. Investigators should remove or quarantine the unlicensed component, then determine whether the remaining evidence constitutes an independently validated assessment. If it does not, the battery-level output remains unlicensed. If it does, the remaining evidence can be classified without pretending that the failed component supplied support.

4. The Indeterminacy Profile

A four-outcome label is still too coarse for cumulative science. Every indeterminate or unlicensed result should be accompanied by a structured cause profile. Five dimensions capture many recurrent problems.

For assessment $\mathcal{A}$, let:

$$P_R(\mathcal{A}) = \langle e, s, m, b, t \rangle,$$

where e records evidential scarcity, s surrogate discordance, m model dependence, b boundary or grain sensitivity, and t transport uncertainty. Each entry may be binary for a simple registry or graded on a preregistered scale for research use.

Plain-language reading. The four-outcome label says where the assessment stopped. The profile says why. Two systems can both receive an indeterminate result while requiring entirely different next studies.

Why this is not a consciousness score. The entries describe limitations in an inference, not dimensions or quantities of experience. They should not be summed into a single “uncertainty number” unless an application validates such an aggregation. In particular, low values do not prove consciousness and high values do not prove its absence.

How the profile should be used. Each nonzero entry must be supported by an auditable reason and paired with a resolution path. If the profile changes across reasonable analytic choices, that instability is itself part of the result. The profile is therefore diagnostic of the assessment process rather than decorative metadata.

4.1 Evidential scarcity

There may be too little reliable evidence to discriminate between presence and absence. The data can be valid yet weak: a score lies between thresholds, confidence intervals span both decision regions, or repeated trials yield high variance. The appropriate response is more or better-powered measurement, not theoretical reinterpretation.

Evidential scarcity must be distinguished from a negative result. Low signal can support absence only when the test’s detection process is understood and the observed signal is lower than would be expected under conscious-

ness. Otherwise, low signal is simply low information.

4.2 Surrogate discordance

Different measures intended to track the same target may conflict. Behavioral examination may indicate unresponsiveness while task-based neuroimaging indicates command following. Complexity measures may remain high while recurrent markers weaken. Two algorithms applied to the same recording may classify it differently.

Discordance can be scientifically productive because it exposes differences in task demand, temporal sensitivity, construct coverage, or confounding. It should be reported rather than averaged away unless the aggregation rule was justified and preregistered. The resolution path is measurement decomposition: determine whether the surrogates truly address the same target, identify their auxiliary demands, and use interventions or repeated measures to adjudicate the conflict.

A preregistered discordance rule can make this constraint explicit. Let $z_1, \dots, z_k$ be normalized outputs from credible surrogates intended to bear on the same target, let d be an appropriate disagreement function, and let κ be the maximum tolerated discordance:

$$G(E) = \max_{j,k} d(z_j, z_k), \quad G(E) > \kappa \Rightarrow D(\mathcal{A}) = \text{Indeterminate.}$$

Plain-language reading. If the strongest disagreement among credible measures exceeds the amount accepted in advance, the battery may not select the convenient measure or conceal the conflict in an average. It returns indeterminate.

Assumptions. The surrogates must be commensurable enough for d to be meaningful, and each must satisfy its own validity conditions. A technically failed measure should be marked unlicensed rather than counted as substantive disagreement. A justified hierarchy may also designate one measure as primary, but that hierarchy must precede the result and must explain why discordant secondary evidence does not overturn classification.

What would revise the rule. If empirical validation shows that particular patterns of disagreement reliably predict consciousness status, those patterns should become part of a richer classifier rather than remaining automatic indeterminacy triggers. Until then, preserving disagreement is epistemically safer than post hoc aggregation.

4.3 Model indeterminacy

Evidence may be clear while its significance depends on a contested theory. A system might display recurrent processing but lack a global workspace; exhibit global availability but lack a higher-order representation; or possess several computational indicators while lacking biological features regarded as necessary by some theories. In such cases the indeterminacy profile should state which theoretical commitments produce which classifications.

This is not a reason to abandon theory-derived tests. It is a reason to avoid presenting a theory-conditional output as theory-independent. Adversarial tests and preregistered comparisons can reduce model indeterminacy by identifying observations on which rival theories make divergent predictions (Cogitate Consortium et al., 2025; Doerig et al., 2021).

Let $\mathcal{M}$ be the preregistered set of live inferential models and define:

$$O_{\mathcal{M}} = \{D(\mathcal{A}_m) : m \in \mathcal{M}\}.$$

Plain-language reading. Apply each scientifically live model to the same bounded system and evidence. If the models agree, the classification is model-robust relative to that set. If they yield different outputs, report the conditional results rather than presenting one as field-wide consensus.

Weighting and scope. The framework does not require all models to receive equal weight. Differential weighting is defensible when based on independently stated empirical support, scope, and predictive performance. It is not defensible when weights are chosen after observing which model gives the desired consciousness result.

Resolution path. Model indeterminacy is reduced by evidence on which the models make discriminating predictions, by improved validation of the indicator-target bridge, or by justified retirement of a model. Mere accumulation of evidence that every model already accommodates will not resolve the disagreement.

4.4 Boundary and grain indeterminacy

Measures of integration, broadcast, complexity, and recurrence depend on which parts of a system are included and over what temporal window. A candidate system may contain nested or partially coupled organizations. Split brains, brain-body loops, organoid-computer interfaces, distributed robots, and modular artificial agents make this problem explicit, but ordinary neuroscience also selects spatial and temporal boundaries.

If reasonable boundary choices reverse the result, the attribution is boundary-sensitive. The appropriate response is not to select the favorable boundary after inspection. Investigators should preregister candidate boundaries, report sensitivity across them, and use perturbations to test causal autonomy. Until the attribution is stable or a principled individuation rule is supplied, the result remains indeterminate or unlicensed.

Let $\mathcal{B}$ be the preregistered set of scientifically defensible boundaries or grains, and let $\mathcal{A}_b$ be the assessment conducted under boundary b . Define the boundary-output set:

$$O_{\mathcal{B}} = \{D(\mathcal{A}_b) : b \in \mathcal{B}\}.$$

The classification is boundary-robust when $|O_{\mathcal{B}}| = 1$. If $|O_{\mathcal{B}}| > 1$, boundary sensitivity must be reported and the field-level result is indeterminate unless an independent individuation rule privileges one boundary.

Plain-language reading. Reanalyze the system using every boundary that was defensible before seeing the outcome. If all choices produce the same classification, the result is robust to system individuation. If they do not, the location or extent of the candidate conscious system remains part of the unresolved question.

Why independence matters. A boundary cannot be privileged merely because it yields the preferred consciousness result. It must be supported by causal autonomy, effective coupling, regulatory closure, perturbational evidence, or another criterion that does not presuppose the classification at issue.

Limitation. Agreement across $\mathcal{B}$ proves robustness only relative to the candidate set. If investigators omit a scientifically plausible boundary, $|O_{\mathcal{B}}| = 1$ can create false reassurance. The candidate set and the reasons for

inclusion must therefore be public.

4.5 Transport indeterminacy

A test validated in one domain may not preserve its meaning in another. Transportability in causal inference concerns the conditions under which findings from one population can license conclusions in another (Pearl & Bareinboim, 2014). Consciousness tests face an analogous but broader problem because the indicator-target relation itself may change. Physiological markers calibrated in adult humans may not transfer directly to infants, nonhuman animals, damaged brains, organoids, or artificial systems. Even within humans, etiology, medication, sensory impairment, and arousal can alter test sensitivity.

Transport indeterminacy is especially important in artificial consciousness. Indicator properties derived from neuroscientific theories offer a disciplined starting point (Butlin et al., 2023), but implementing an indicator is not equivalent to validating its evidential force in a new substrate. Structural or computational analogy can generate a research hypothesis without yet generating a licensed consciousness classification.

Let P_0 be the population or substrate in which test T was validated and P_1 the target domain. A transport claim can be represented as:

$$\text{Tr}(T : P_0 \rightarrow P_1) = 1,$$

only when the evidence-indicator relation, the indicator-target rationale, and the relevant detection conditions are sufficiently preserved. If transport has not been established, applying T in P_1 may yield exploratory evidence but not a licensed positive or negative classification.

Plain-language reading. A test does not carry its meaning merely because its code, formula, or sensor can be run on a new system. Investigators must justify why the resulting value bears the same evidential relation to consciousness in the new domain.

Three bridge requirements. Construct preservation asks whether the test still measures the same functional or dynamical property. Causal-role preservation asks whether that property plays the same theoretically relevant role. Calibration preservation asks whether thresholds and error rates remain defensible. Failure at any bridge can produce transport indeterminacy or render the application unlicensed.

What would establish transport. Evidence may include successful prediction in intermediate populations, mechanistic correspondence, measurement invariance, interventions showing preserved causal role, and convergence with independently motivated indicators. Surface similarity alone is insufficient, but exact material identity is not assumed.

4.6 A reporting template

A compact report could take the following form:

Field	Required entry
Target	Exact form of consciousness being assessed
Candidate system	Boundary, grain, and time window

Field	Required entry
Test and model	Assessment battery and inferential assumptions
Validity envelope	Populations, substrates, and conditions covered
Outcome	Positive, negative, indeterminate, or unlicensed
Cause profile	Evidential, surrogate, model, boundary, and transport status
Robustness	Sensitivity to thresholds, preprocessing, and boundaries
Resolution path	Evidence or intervention most likely to change the result
Decision note	Practical action considered separately from scientific status

The resolution path is essential. A useful indeterminate result narrows the next question. “More research is needed” is not enough.

5. Worked Applications

5.1 Disorders of consciousness

Suppose a patient receives no diagnosis of command following at the bedside, but EEG data show a weak task-dependent response that does not cross a preregistered threshold. If the EEG recording is adequate and the task is appropriate, the result is indeterminate, not negative. Evidential scarcity is primary. If the patient has fluctuating arousal and the protocol was administered only once, the validity of a negative inference is further weakened; repeated standardized assessment is indicated (Giacino et al., 2018; Wannez et al., 2017).

Now suppose behavioral assessment repeatedly indicates unresponsive wakefulness while fMRI reveals reproducible command-following imagery. The result is not a simple disagreement to be averaged. The behavioral test may be insensitive to covert command following because it requires motor output. For preserved command comprehension and volitional imagery, the evidence supports a positive attribution of at least some awareness-relevant capacity. Questions about ongoing phenomenal richness or continuity may remain indeterminate. The framework permits target-specific positive and indeterminate results to coexist.

Finally, suppose no task-based response is detected, but the patient cannot reliably hear instructions and sedating medication is present. The result is unlicensed for denial of consciousness. The protocol failed to establish the auxiliary conditions needed for a valid negative inference.

5.2 Nonhuman animals

Animal consciousness research often relies on clusters of behavioral, cognitive, anatomical, and physiological indicators rather than a decisive test. Theory-light approaches seek capacities that conscious perception plausibly facilitates while avoiding commitment to a single full theory (Birch, 2022). A multidimensional profile may also be more informative than ranking species on one consciousness scale (Birch et al., 2020).

Suppose an animal displays flexible learning, motivational tradeoffs, and protective behavior after injury, but the relevant neuroanatomy differs from that of mammals. The result may support sentience under a theory-light indicator framework while remaining transport-indeterminate for a mammal-calibrated neural marker. The proper report should preserve both facts. Failure to instantiate the mammalian mechanism is not automatically evidence of absence if functional alternatives remain plausible.

Conversely, a single behavior that can be generated by a simple reflexive mechanism may be insufficient to discriminate conscious from nonconscious processing. That is a valid indeterminate result if the behavior bears some relation to the target but lacks specificity. It should motivate tests designed to exclude the simpler explanation.

5.3 Neural organoids

Current organoids create an acute transport problem. Neural activity, oscillation, complexity, or perturbational responsiveness can be measured, but the interpretation of such measures was developed in systems with sensory histories, differentiated anatomy, vascular and regulatory support, and behavior. A human-like numerical value does not by itself establish equivalence of causal role.

If an organoid exhibits a proposed consciousness marker but lacks an established system boundary or the supporting organization required by the marker's theory, the consciousness result is unlicensed rather than positive. This does not prove absence. It states that the inferential bridge has not been validated. Research should focus on whether the marker is mechanistically homologous, whether relevant causal interactions exist, and whether the candidate system possesses the enabling conditions presupposed by the assessment.

5.4 Artificial systems

Artificial systems can produce fluent self-reports, implement recurrent computation, maintain global workspaces, or satisfy other theory-derived indicators. These features may be architecturally real even when their interpretation is disputed. The four-outcome framework blocks two opposite errors: dismissing every artificial system because it lacks biological markers, and declaring consciousness whenever a computational analogue is identified.

An assessment of AI consciousness should report indicator evidence and transport status separately. If multiple theory-derived properties are credibly implemented but no cross-substrate calibration exists, the result may be model- and transport-indeterminate. If a test depends on biological processes absent from the system, its application is unlicensed rather than negative. If a theory explicitly predicts substrate-independent sufficiency and its specified conditions are met, the output can be positive relative to that theory while remaining model-indeterminate at the field level.

This conditional reporting is not evasive. It identifies precisely where disagreement lies: in implementation facts, substrate requirements, or the truth of the theory.

5.5 The N* model as an example

The N* model proposes that minimal phenomenal presence obtains exactly when a viable system instantiates integration or synergy, broadcast availability, and recurrent stability at the relevant grain (Stilwell, 2026). Its measurement policy already allows indeterminate classification when preregistered surrogates conflict. The present framework generalizes and sharpens that policy.

Under the four-outcome scheme, a system whose valid measures of integration disagree beyond tolerance is indeterminate. A system whose viability has not been assessed, whose graph boundary is selected retrospectively, or whose measures have not been validated for its substrate is unlicensed. A system is negative only when the

full assessment is applicable and one or more proposed necessary conditions are robustly absent. This distinction protects the model from neither falsification nor counterexample; it specifies when an apparent counterexample is evidentially mature enough to bear that role.

6. Indeterminacy and Scientific Progress

6.1 Indeterminacy is informative when it is structured

A critic may object that science should reduce uncertainty, not institutionalize it. But an explicit indeterminate category can accelerate reduction of uncertainty by preventing heterogeneous failures from being pooled. The scientific value lies in decomposition. Evidential scarcity calls for more observations. Surrogate discordance calls for construct analysis and adjudicating interventions. Model indeterminacy calls for discriminating predictions. Boundary indeterminacy calls for causal individuation. Transport indeterminacy calls for bridge validation.

This converts a terminal statement - “we do not know” - into a research map. A mature field does not prove its rigor by forcing every measurement into a binary output. It proves rigor by matching the strength of its conclusions to the strength and domain of its evidence.

6.2 Indeterminate rates are properties of tests

The frequency of indeterminate and unlicensed outputs should be reported as part of test performance. A test that classifies only easy cases with high apparent accuracy may have limited usefulness if it returns no interpretable result in the population where it is most needed. Conversely, a test with a wide indeterminate region may be appropriately calibrated when false classification is costly.

At minimum, validation studies should report:

- the proportions of positive, negative, indeterminate, and unlicensed outputs;
- the reasons for indeterminate and unlicensed classifications;
- performance estimates under transparent treatments of inconclusive cases;
- robustness across thresholds and repeated measurements;
- the population and substrate boundaries of the validity envelope.

Three summary quantities should accompany ordinary accuracy estimates. If N_+ , N_- , N_I , and N_U are the numbers of positive, negative, indeterminate, and unlicensed outputs, and $N = N_+ + N_- + N_I + N_U$ includes every attempted assessment in the stated study cohort, then:

$$\text{Coverage} = \frac{N_+ + N_-}{N}, \quad r_I = \frac{N_I}{N}, \quad r_U = \frac{N_U}{N}.$$

When an independent reference classification is available, the error rate among cases the test actually classifies should also be reported:

$$R_{\text{sel}} = \Pr(\text{classification error} \mid D \in \{\text{Positive, Negative}\}).$$

Plain-language reading. Coverage says how often the procedure makes a polarity classification. The indeterminate and unlicensed rates say why it did not classify the remainder. Selective risk says how often the procedure is wrong among the cases it chose to classify.

Why this matters. A test can appear highly accurate by classifying only easy cases, or appear comprehensive by forcing hard and invalid cases into positive or negative bins. Reporting coverage together with selective risk and the two unresolved rates exposes that tradeoff, a central concern in classification with a reject option (Chow, 1970).

Limitation. A trustworthy reference classification is often unavailable in consciousness science. In that setting R_{sel} may not be estimable without circularity. Coverage, r_I , and r_U remain reportable, while accuracy claims should be restricted to validation contexts where the reference standard is independently defensible.

Excluding indeterminate results from accuracy calculations can make a test appear more decisive than it is (Shinkins et al., 2013). Coding them as negative can conceal sensitivity failures. Coding unlicensed assessments as indeterminate can conceal non-applicability. Consistent with the transparency goals of STARD, validation reports should preserve participant flow, exclusions, missingness, and every output category rather than presenting accuracy only among successfully classified cases (Bossuyt et al., 2015). All four outcomes matter, and the denominator used for each performance statistic must be explicit.

6.3 Preregistration prevents opportunistic certainty

Thresholds, surrogate hierarchies, conflict rules, exclusion criteria, and system boundaries should be set before the critical result is examined. Otherwise, investigators can resolve ambiguity opportunistically by selecting a preferred measure, time window, preprocessing pipeline, or candidate boundary. Consciousness research is particularly vulnerable because many plausible measures exist and theories permit different post hoc interpretations (Yaron et al., 2022).

Preregistration cannot validate a poor test, but it can reveal whether certainty was earned by the protocol or manufactured after inspection.

6.4 Resolution is source-specific

Indeterminate and unlicensed results are not resolved in the same way. Let E' represent additional evidence and $\mathcal{V}'$ an expanded or repaired validity envelope. The characteristic transitions are:

$$\text{Indeterminate} \xrightarrow{E'} \{\text{Positive, Negative, Indeterminate}\},$$

whereas:

$$\text{Unlicensed} \xrightarrow{\mathcal{V}'} \{\text{Positive, Negative, Indeterminate, Unlicensed}\}.$$

Plain-language reading. A valid but nondiscriminating test may be resolved by more informative evidence. An unlicensed assessment first requires repair of applicability, measurement quality, target specification, boundary choice, or transport. Merely collecting more data with the same invalid procedure does not solve the problem.

Why this matters. The distinction prevents wasteful repetition. Low statistical power calls for more observations; a broken inferential bridge calls for a different study. In the latter case, additional precision can make an invalid answer more reproducible without making it more meaningful.

Success condition. A resolution is scientifically successful only when the new evidence or validity work addresses the recorded cause profile and the resulting classification is robust under preregistered sensitivity analysis. Changing the label without removing the stated source of indeterminacy is not resolution.

7. From Scientific Status to Practical Action

Scientific classification and practical decision are related but distinct. A positive consciousness result normally strengthens reasons for protection. A negative result may weaken them. An indeterminate result, however, does not dictate either action or inaction. Decisions also depend on the possible magnitude of harm, reversibility, available alternatives, and the asymmetric costs of false attribution and false denial.

Let the scientific assessment produce an evidential state D , while a decision rule selects an action A by considering D together with a loss structure Λ :

$$A^* = \arg \min_A \mathbb{E}[\Lambda(A, C) \mid D].$$

Plain-language reading. Choose the available action with the lowest expected loss, given the scientific result and the remaining uncertainty about consciousness. The loss function includes the costs of harming a conscious system, withholding benefits, imposing unnecessary restrictions, delaying research, and making reversible or irreversible errors.

Why this matters. Two institutions can accept the same indeterminate scientific result yet choose different actions because the stakes differ. Continuing a reversible, non-invasive measurement protocol is not equivalent to withdrawing life-sustaining treatment, exposing an animal to severe harm, scaling an organoid architecture, or deploying an artificial agent under conditions it describes as aversive. Precaution should be proportionate to both evidence and possible harm (Birch, 2024).

Assumptions and limitations. The expression does not assume that every value can be reduced to money or that precise probabilities are available. It represents the structure of a decision: actions must be compared under uncertainty and asymmetric error costs. Institutions should disclose qualitative judgments, probability ranges, and protected values rather than hiding them inside an apparently objective number.

What the formula does not license. A precautionary action does not upgrade D from indeterminate to positive, and a permissive action does not downgrade it to negative. New scientific classification requires new evidence, improved validity, or a justified change in the inferential model. Decision urgency cannot substitute for epistemic support.

The separation also protects scientific language. Ethical precaution need not redescribe an indeterminate system as conscious. One can adopt safeguards because consciousness remains a credible possibility. Conversely, scientific indeterminacy should not be invoked as a universal veto on research. The appropriate response can range from documentation and monitoring to redesign, independent review, or temporary restriction.

8. Objections and Replies

Objection 1: The framework merely renames uncertainty

The proposal would be trivial if it added only a third label. Its substantive content is the distinction between indeterminate and unlicensed outputs, the Negative License Principle, the validity envelope, the five-part cause profile, and the requirement that each unresolved result specify a resolution path. These constraints alter test validation, reporting, and inference.

Objection 2: Four outcomes weaken falsifiability

Any theory could be protected if every adverse result were called indeterminate. The remedy is preregistration. Applicability conditions, thresholds, conflict rules, and evidence required for positive and negative classification must be specified before outcomes are known. A result that falls in a preregistered negative region counts against a theory that predicted consciousness. Indeterminacy should not be available as a post hoc rescue category.

Objection 3: Consciousness is either present or absent

The framework does not deny bivalence at the level of reality. It distinguishes the state of the system from the state of warranted classification. A disease may be present or absent while a diagnostic result remains inconclusive. The same logical distinction applies to consciousness even if consciousness is metaphysically binary.

Objection 4: Probabilities already capture uncertainty

Probabilities can represent degrees of confidence but do not by themselves identify why confidence is limited or whether the test applies. A precise posterior produced by an invalid transport assumption remains unlicensed. Two cases can receive the same posterior for different reasons and require different next experiments. The cause profile supplements rather than competes with probabilistic analysis.

Objection 5: Model disagreement makes every hard case indeterminate

Not every disagreement is equally relevant. Some evidence is robust across theories, and some theories are better supported than others. The framework does not require unanimity. It requires that theory-conditional conclusions be labeled as such and that field-level conclusions reflect material, live disagreement. As theories gain or lose support, model indeterminacy can shrink.

Objection 6: The unlicensed category is too severe

“Unlicensed” does not mean scientifically worthless. An out-of-domain result may generate a valuable hypothesis or motivate calibration. It means only that the result does not yet warrant the intended consciousness attribution. This vocabulary protects exploratory evidence from being overinterpreted while preserving its role in discovery.

Objection 7: Precaution will bias the science toward positive attribution

Scientific outputs should not be altered to obtain preferred ethical consequences. That is why classification and action are separated. An indeterminate result remains indeterminate even when precautionary protection is justified. The framework is symmetrical at the scientific level and explicitly loss-sensitive at the decision level.

Objection 8: The framework is only a reject option with new terminology

Reject-option and three-way classifiers are important predecessors, and the paper now makes that debt explicit. Their ordinary boundary or abstention output corresponds to valid nondiscrimination. The additional contribution is to separate that state from failure of inferential license. A patient near a validated threshold and an artificial system outside the test's validation domain should not receive the same scientific description merely because neither receives a polarity classification.

Objection 9: The framework is too burdensome for routine use

Full cause profiles and sensitivity analyses impose costs. But the highest burden is appropriate in cases where classification is uncertain and consequential. Routine use can employ the compact protocol in Appendix B and escalate only when an output is indeterminate, unlicensed, boundary-sensitive, or decision-critical. The framework is modular: the minimum is explicit target, validity check, data-quality check, four-output classification, and reason code.

9. Recommendations

Researchers developing or applying consciousness tests should adopt the following minimum practices:

1. Specify the target form of consciousness rather than using “consciousness” without qualification.
2. Define the candidate system, spatial grain, temporal window, and relevant auxiliary capacities.
3. State the test's validity envelope and identify all extrapolations beyond it.
4. Pre-register rule-in and rule-out thresholds, leaving an explicit indeterminate region where justified.
5. Pre-register positive- and negative-license conditions, including relevant confounds and auxiliary demands.
6. Give validity and data-quality failure precedence over evidential thresholds.
7. Separate valid indeterminate results from technically invalid or interpretively unlicensed assessments.
8. Report disagreements among surrogates before applying any composite score.
9. Attach an indeterminacy profile covering evidence, surrogates, models, boundaries, and transport.
10. State the observation or intervention most likely to resolve each identified source of indeterminacy.
11. Report coverage, indeterminate rate, unlicensed rate, exclusions, and denominators in validation studies.
12. Keep scientific attribution separate from ethical or clinical action while making the decision rule and error costs explicit.

These practices are compatible with binary, graded, multidimensional, theory-heavy, and theory-light models of consciousness. They regulate the inference from evidence to attribution rather than dictating the correct theory of consciousness.

10. Conclusion

At the boundaries of consciousness science, forced binary classification creates an illusion of knowledge. A failed response becomes absence of experience; a conflicting biomarker is averaged away; an out-of-domain measure is treated as if its meaning traveled with its numerical value. These errors are avoidable.

Consciousness assessments should return four outcomes: positive, negative, indeterminate, and unlicensed. Negative attribution requires more than failure to detect a positive marker. Indeterminate results require a cause profile. Unlicensed results must be separated from valid gray-zone evidence. Each unresolved classification should identify a concrete resolution path.

The central claim is methodological but not modest in consequence: indeterminacy can be a successful scientific result. Properly structured, it identifies the exact limit of an inference, prevents false certainty, and tells investigators what kind of evidence would move the question forward. In a field without a universal gold standard and with increasingly consequential applications, knowing why a conclusion is not yet licensed is part of knowing what the evidence shows.

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Appendix A: Glossary and notation

This appendix consolidates the framework’s technical vocabulary. The final column records the inference that each item is designed to block.

Term or symbol	Role in the framework	Interpretive guardrail
S	Candidate system at a specified boundary and grain	Do not assume the relevant system is the whole organism, cortex, device, or network without argument.
C	Target form of consciousness	Specify phenomenal presence, conscious content, sentience, wakeful awareness, or another target; do not treat them as interchangeable.
T	Test or assessment battery	A test is an evidential procedure, not the target itself.
E	Evidence produced by the assessment	Evidence can be precise yet weak, conflicted, or misinterpreted.
M	Inferential model connecting evidence to target	A theory-conditional result is not automatically a field-level result.
$\mathcal{V}$	Validity envelope	State the populations, substrates, tasks, boundaries, and conditions over which interpretation is supported.
$\mathcal{A}$	Full assessment tuple $\langle S, C, T, E, M, \mathcal{V} \rangle$	A consciousness attribution is indexed to all six elements, not to a score alone.
$a(\mathcal{A})$	Applicability gate	Failure means the intended inference is unlicensed, not negative.
$q(\mathcal{A})$	Execution and data-quality gate	Technical failure must not be redescribed as target absence.
$c(E, M)$	Unresolved surrogate-conflict flag	Credible discordance takes precedence over composite scoring.
$L(E; T, C, M)$	Validated evidential-support score or grade	It is not an amount of consciousness.
τ_-	Rule-out threshold	Crossing it licenses negative evidence only under the Negative License Principle.
τ_+	Rule-in threshold	Crossing it licenses positive evidence only under the Positive License Principle.
$D(\mathcal{A})$	Four-output classification	Positive, negative, indeterminate, or unlicensed; these are not four ontological states.
$P_R(\mathcal{A})$	Reason or cause profile $\langle e, s, m, b, t \rangle$	The entries diagnose inferential limitations and should not be summed into a consciousness score.
κ	Preregistered surrogate-discordance tolerance	It cannot be adjusted after seeing which measure favors the desired result.
$\mathcal{M}$	Set of live inferential models	The set and any weights must be justified independently of case outcome.

Term or symbol	Role in the framework	Interpretive guardrail
$\mathcal{B}$	Set of defensible system boundaries or grains	Robustness is only as good as the completeness of this candidate set.
Tr	Transport-license claim	Running a test in a new domain is not equivalent to validating its interpretation there.
R_{sel}	Error rate among polarity-classified cases	Report together with coverage; low error achieved by classifying very few cases can be misleading.
Λ	Decision loss structure	Practical values and error costs affect action, not the scientific classification itself.

Appendix A.1 Outcome vocabulary

Positive means that an applicable, adequately executed, nonconflicted assessment crosses a preregistered rule-in standard under an explicit model.

Negative means that an applicable, adequately executed, nonconflicted assessment crosses a preregistered rule-out standard under an explicit model.

Indeterminate means that the assessment is licensed and informative but does not discriminate sufficiently, either because evidence lies in the gray zone or because credible surrogates or models remain unresolved.

Unlicensed means that the procedure cannot support the intended interpretation in the present case. It is a meta-output about the assessment, not evidence that consciousness is absent.

Appendix B: Classification protocol and precedence rules

The protocol below is designed for preregistration and independent replication. Later stages should not be used to repair failures at earlier stages.

Stage	Required question	If the requirement passes	If it fails
1. Specify target	What exact form of consciousness is being assessed?	Record C .	Unlicensed until the target is specified.
2. Bound system	What nodes, processes, body components, devices, and time window constitute S ?	Record primary and sensitivity boundaries.	Unlicensed if no defensible candidate system is available.
3. State model	Why should evidence from T bear on C ?	Record M and theory-conditional predictions.	Unlicensed if no indicator-target rationale is supplied.
4. Audit validity	Is the case within $\mathcal{V}$, or has transport been independently justified?	Set $a = 1$.	Set $a = 0$: unlicensed.
5. Audit execution	Were auxiliary demands, protocol fidelity, and data quality adequate?	Set $q = 1$.	Set $q = 0$: unlicensed.
6. Test conflict	Do credible surrogates disagree beyond κ ?	Set $c = 0$ and continue.	Set $c = 1$: indeterminate.
7. Apply thresholds	Where does L fall relative to τ_- and τ_+ ?	Positive, negative, or gray-zone indeterminate.	Do not invent a post hoc cutoff.
8. Stress-test	Does the output survive defensible models, boundaries, preprocessing, and thresholds?	Mark robust relative to tested set.	Mark indeterminate or qualified by the unstable dimension.
9. Attach profile	Why is any part of the output unresolved?	Record P_R and resolution path.	Report is incomplete.
10. Separate action	What should be done given evidence, stakes, and reversibility?	Record decision analysis separately.	Do not convert urgency into scientific certainty.

Appendix B.1 Compact precedence algorithm

The classification logic can be implemented as the following ordered procedure:

1. If target, system, model, or validity envelope is missing, return **Unlicensed**.
2. If applicability or execution quality fails, return **Unlicensed**.
3. If credible surrogates conflict beyond the preregistered tolerance, return **Indeterminate**.
4. If a validated support score is at or above τ_+ , return **Positive**.
5. If it is at or below τ_- , return **Negative**.
6. Otherwise return **Indeterminate**.
7. In every case, report model, boundary, threshold, and transport sensitivity.

Why order matters. A high score cannot rescue an invalid application, and a low score cannot turn technical failure into evidence of absence. Likewise, averaging occurs only after conflict has been assessed. This order prevents downstream calculations from laundering upstream inferential failures.

Appendix B.2 Formal consistency requirements

For a fully specified assessment with binary gates and ordered thresholds, the rule should satisfy:

- **Exhaustivity:** every assessment returns one output.
- **Mutual exclusivity:** no assessment simultaneously receives two outputs at the same reporting level.
- **Validity precedence:** $a = 0$ or $q = 0$ always implies unlicensed, regardless of L .
- **Conflict precedence:** when $a = q = 1$, unresolved $c = 1$ implies indeterminate before aggregation.
- **Threshold ordering:** $\tau_- < \tau_+$, preserving a nonempty gray zone when the evidence warrants one.
- **Locality:** the output applies only to the stated S , C , M , and $\mathcal{V}$.
- **Revision transparency:** a changed output must identify the new evidence, validity repair, or model change responsible.

These are coherence conditions on the reporting framework. They do not establish that a particular C-test is valid.

Appendix C: Preregistration and reporting checklist

Appendix C.1 Before collecting or inspecting critical data

- ☐ Define the target form of consciousness and distinguish it from report, intelligence, arousal, metacognition, and welfare capacity where relevant.
- ☐ Define the primary candidate system, spatial grain, temporal window, and content or state under assessment.
- ☐ List alternative defensible system boundaries for sensitivity analysis.
- ☐ Name the inferential model or models and state the indicator-target rationale for each.
- ☐ State the validation population, task, substrate, and measurement conditions.
- ☐ Identify every extrapolation beyond the established validity envelope.
- ☐ Specify auxiliary demands such as comprehension, attention, memory, sensory access, motor capacity, or regulatory support.
- ☐ Select primary and secondary surrogate measures before outcome inspection.
- ☐ Define preprocessing, artifact rejection, missing-data handling, and minimum data-quality rules.
- ☐ Pre-register τ_- , τ_+ , κ , and any model or surrogate weights.
- ☐ Pre-register positive- and negative-license conditions and known alternative explanations.
- ☐ State what result would count against the test or theory rather than merely producing indeterminacy.

Appendix C.2 During classification

- ☐ Record failures at the stage where they occur; do not carry an invalid score forward.
- ☐ Preserve individual surrogate outputs before computing a composite.
- ☐ Record whether an unlicensed component can be removed without invalidating the remaining battery.
- ☐ Apply the precedence algorithm without outcome-dependent exceptions.
- ☐ Generate the cause profile and name one resolution path for every nonzero entry.
- ☐ Run model, boundary, threshold, preprocessing, and temporal-window sensitivity analyses.
- ☐ Distinguish confirmatory classifications from exploratory out-of-envelope observations.

Appendix C.3 In the final report

- ☐ Report all four output counts and percentages.
- ☐ Report coverage, indeterminate rate, unlicensed rate, exclusions, missingness, and denominators.
- ☐ Report selective error only where an independently defensible reference classification exists.
- ☐ State which results are model-conditional, boundary-sensitive, or transport-dependent.
- ☐ Publish the decision rule, conflict rule, and deviations from preregistration.
- ☐ Explain how each indeterminate or unlicensed result could be resolved.
- ☐ Keep ethical, clinical, or governance recommendations in a section distinct from scientific classification.
- ☐ Make clear that absence of a licensed classification is not evidence of absence.

Appendix D: Worked classification ledger

The ledger illustrates how superficially similar uncertainty can produce different outputs. The values are schematic and are not proposed as empirical thresholds.

1. **Convergent rule-in case.** An adult no-report visual paradigm uses validated measures that converge above τ_+ . Gate state: $a = 1, q = 1, c = 0$. Output: **Positive** for the specified visual-consciousness target. Next step: test robustness across report and no-report conditions rather than generalizing to unrelated targets.
2. **Convergent rule-out case.** A repeated disorder-of-consciousness battery has intact auxiliary conditions and convergent evidence below τ_- . Gate state: $a = 1, q = 1, c = 0$. Output: **Negative** for the preregistered target and window. Next step: repeat across clinically justified windows and preserve the distinction between the tested capacity and consciousness more broadly.
3. **Valid gray-zone case.** An adequately recorded EEG result lies between validated thresholds. Gate state: $a = 1, q = 1, c = 0$. Output: **Indeterminate**. Cause profile: evidential scarcity. Next step: increase informative trials or add an independent measure.
4. **Surrogate-conflict case.** Behavioral assessment shows non-response while task-based neuroimaging reproducibly indicates command following. Gate state: $a = 1, q = 1, c = 1$. Output: **Indeterminate** at a broad target level, with a possible capacity-specific positive result. Next step: decompose targets and auxiliary demands rather than average the measures.
5. **Failed auxiliary-condition case.** No task response is detected while hearing and arousal remain unverified. Gate state: $a = 1, q = 0$; conflict and thresholds are not reached. Output: **Unlicensed**. Next step: repair the administration conditions and repeat. The initial non-response is not negative evidence.
6. **Cross-species transport case.** A mammal-calibrated neural marker is applied to an octopus without bridge validation. Gate state: $a = 0, q = 1$. Output: **Unlicensed** despite a numerical score. Next step: establish construct and causal-role preservation or use species-appropriate indicators.
7. **Cross-substrate transport case.** A human neural complexity measure is mapped to an artificial architecture. Gate state: $a = 0, q = 1$. Output: **Unlicensed** for field-level attribution. Next step: treat the analogue as exploratory and validate the cross-substrate mapping, calibration, and theory dependence.
8. **Model-disagreement case.** An artificial system satisfies several computational indicators under one theory but not another. Applicability holds only relative to the respective models. Output: **Model-indeterminate**, accompanied by conditional positive and negative assessments. Next step: report those conditional outputs and design a discriminating test.
9. **Boundary-sensitive case.** Classification changes across defensible brain-only and brain-body boundaries. Gate state within each boundary: $a = 1, q = 1, c = 0$. Field-level output: **Boundary-indeterminate**. Next step: use causal perturbation and autonomy criteria to privilege a boundary independently of the desired result.

Appendix D.1 Target-specific decomposition

A single case can receive different licensed outputs for different targets. For example, reproducible command-following imagery may support a positive attribution of command comprehension and volitional task performance while leaving continuous phenomenal richness indeterminate. This is not inconsistency because the assessment tuples contain different C terms.

The ledger should therefore be repeated when investigators move from one target to another. A positive result for sentience does not automatically establish self-consciousness; a negative result for reportability does not establish absence of phenomenal presence; and an unlicensed test of one capacity does not erase independent evidence about another.

Appendix E: Falsification, audit, and misuse guardrails

Appendix E.1 How the framework itself can fail

The proposal should not be treated as immune to evaluation. It would require revision if any of the following patterns were established:

1. **No practical separation:** independent reviewers cannot reliably distinguish indeterminate from unlicensed cases even with complete protocols.
2. **No error benefit:** preserving the two categories does not improve calibration, error detection, or interpretive accuracy relative to a simpler three-way framework.
3. **No resolution benefit:** cause profiles do not predict which follow-up studies resolve cases or reduce wasted repetition.
4. **Systematic safe-case concealment:** the indeterminate category persistently absorbs cases that validated evidence can classify reliably.
5. **Systematic rescue behavior:** investigators invoke applicability, conflict, or transport concerns only after results threaten a favored theory.
6. **Unmanageable burden:** the protocol's costs exceed its epistemic benefit even in high-stakes or cross-domain applications.

These possibilities can be studied by inter-rater reliability tests, retrospective reanalysis of C-test datasets, prospective preregistration, and comparisons of error, coverage, and resolution time.

Appendix E.2 Common misuse patterns

Misuse	Why it is invalid	Required correction
"The test was not positive, therefore the system is not conscious."	Collapses non-detection into negative evidence.	Apply the Negative License Principle and classify gray-zone or failed cases appropriately.
"The signal is complex, therefore consciousness is present."	Treats an indicator as identical to the target and ignores specificity.	Apply the Positive License Principle and test alternative explanations.

Misuse	Why it is invalid	Required correction
“The measures disagree, so we averaged them.”	Can conceal material surrogate conflict.	Apply the preregistered discordance rule before aggregation.
“The same algorithm ran successfully on the new substrate.”	Computational executability does not establish transport validity.	Validate construct, causal-role, and calibration preservation.
“The preferred boundary gives a stable result.”	Selects system individuation using the desired outcome.	Test all preregistered defensible boundaries or justify one independently.
“The outcome is indeterminate, so no protection is justified.”	Confuses scientific status with practical action.	Conduct a separate loss- and precaution-sensitive decision analysis.
“Precaution requires calling the system conscious.”	Alters scientific language to obtain an ethical result.	Preserve indeterminacy while adopting proportionate safeguards if warranted.
“Unlicensed means evidence of absence.”	Treats failure of inference as a polarity verdict.	Report that the present assessment cannot answer the target question.

Appendix E.3 Minimal audit record

Every published classification should make the following compact record recoverable:

Target - System - Test - Model - Validity - Quality - Conflict - Threshold position - Output - Cause profile - Robustness - Resolution path - Decision note

If any field is missing, readers should be able to identify exactly which inference cannot be audited. This record is the minimum bridge between the conceptual framework and cumulative empirical practice.